Chapter 2
Introduction to Transaction Processing



Accounting Information Systems, 7e
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Objectives for Chapter 2

- Understand the broad objectives of transaction cycles.
- Recognize the types of transactions processed by each of the three transaction cycles
- Know the basic accounting records used in TPS.
- Understand the relationship between the traditional accounting records and their magnetic equivalents.
- Be familiar with documentation techniques.
- Understand the differences between batch and real-time processing and the impact of these technologies on transaction processing.
- Be familiar with data coding schemes used in AIS.

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A Financial Transaction is...

- an economic event that affects the assets and equities of the firm, is reflected in its accounts, and is measured in monetary terms.
- similar types of transactions are grouped together into **three transaction cycles:**
  - the expenditure cycle
  - the conversion cycle
  - the revenue cycle

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Relationship between Transaction Cycles

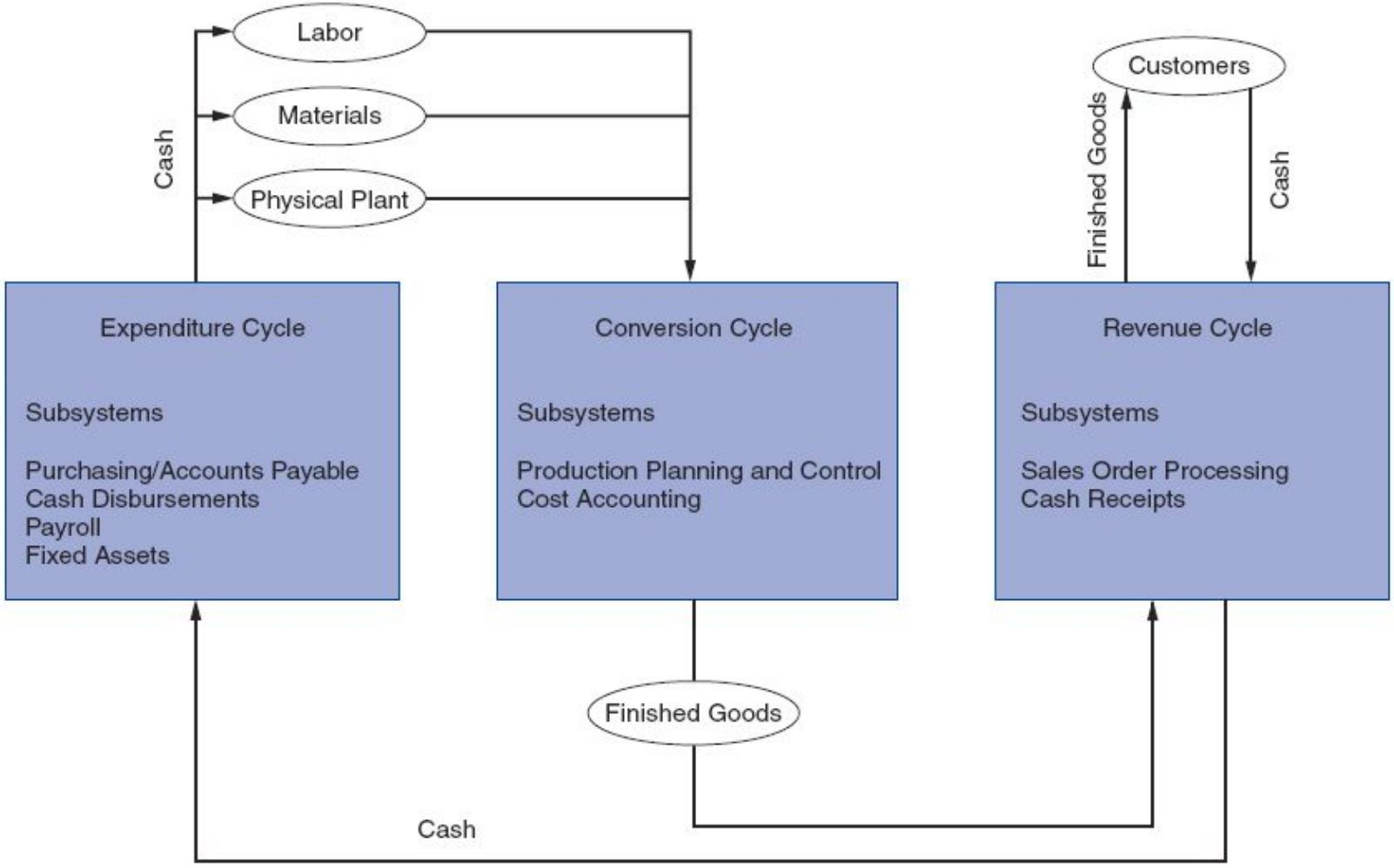


Figure 2-1

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Each Cycle has Two Primary Subsystems

- Expenditure Cycle: time lag between the two due to credit relations with suppliers:
  - **physical component** (acquisition of goods)
  - **financial component** (cash disbursements to the supplier)
- Conversion Cycle :
  - **the production system** (planning, scheduling, and control of the physical product through the manufacturing process)
  - **the cost accounting system** (monitors the flow of cost information related to production)
- Revenue Cycle: time lag between the two due to credit relations with customers :
  - **physical component** (sales order processing)
  - **financial component** (cash receipts)

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Manual System Accounting Records

- **Source Documents** - used to capture and formalize transaction data needed for transaction processing
- **Product Documents** - the result of transaction processing
- **Turnaround Documents** - a product document of one system that becomes a source document for another system

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Manual System Accounting Records

- **Journals** - a record of chronological entry
  - **special journals** - specific classes of transactions that occur in high frequency
  - **general journal** - nonrecurring, infrequent, and dissimilar transactions
- **Ledger** - a book of financial accounts
  - **general ledger** - shows activity for each account listed on the chart of accounts
  - **subsidiary ledger** - shows activity by detail for each account type

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Flow of Information from Economic Event Into the General Ledger

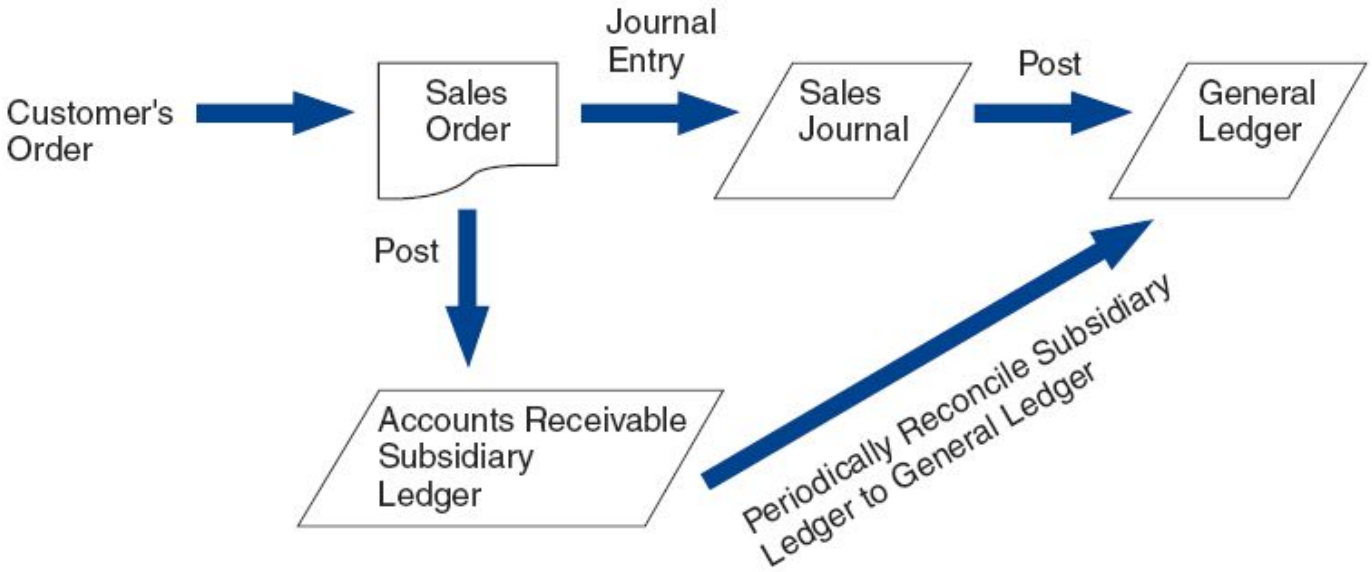


Figure 2-8

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Accounting Records in a Computer-Based System

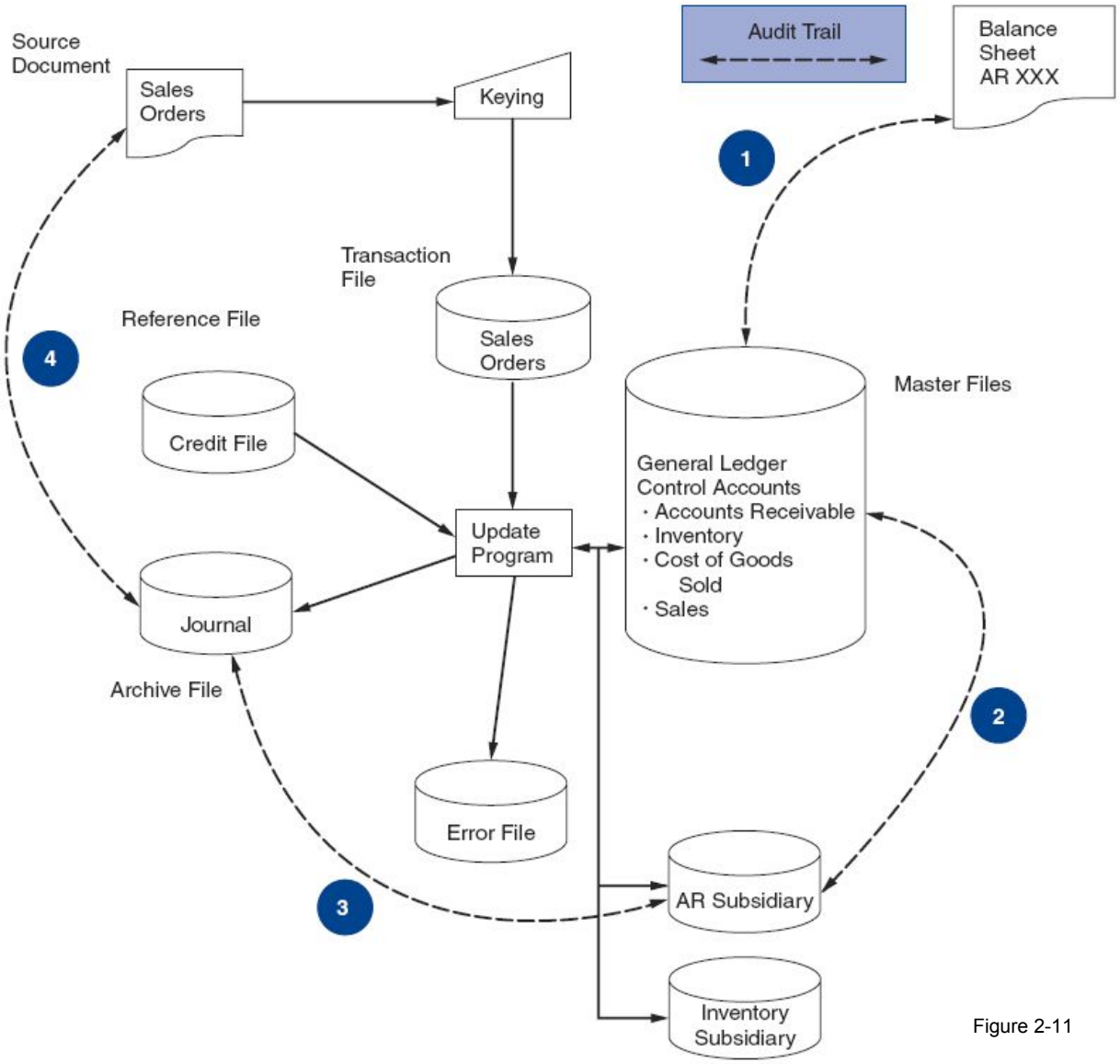


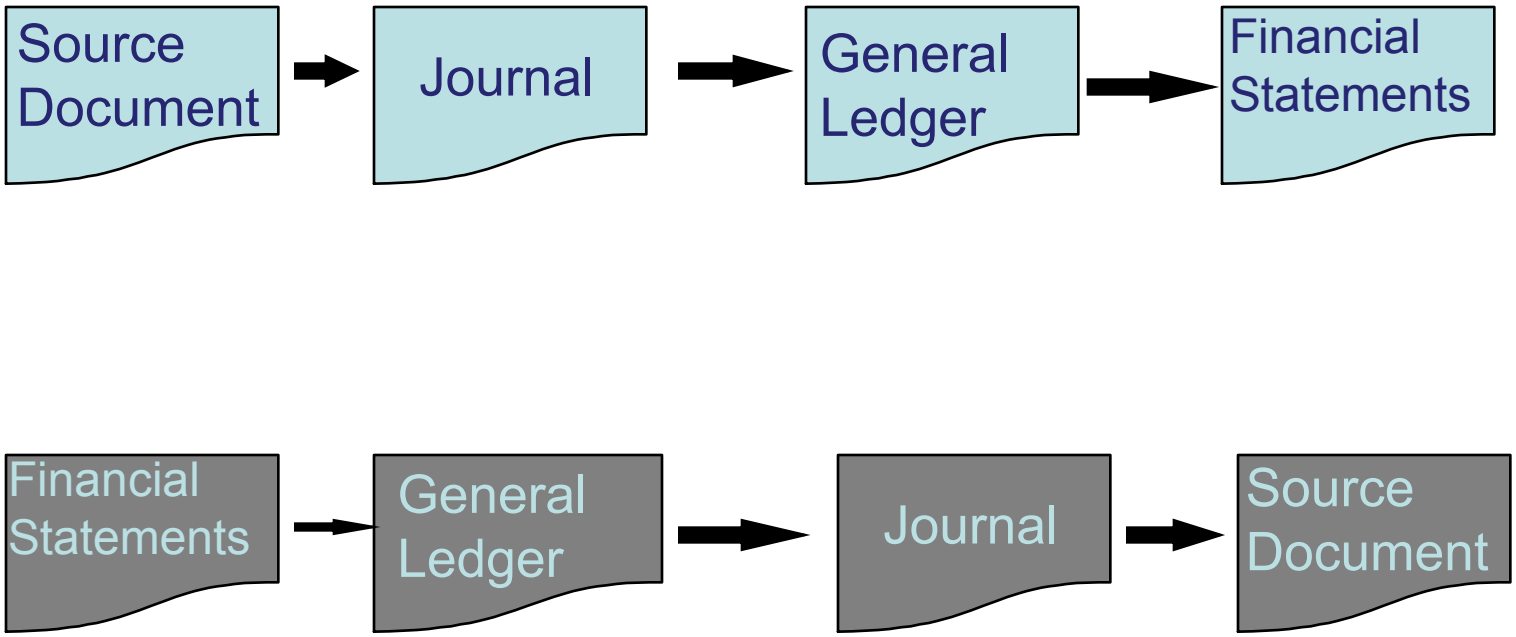
Figure 2-11

EXPLANATION OF STEPS IN FIGURE:

1. Compare the AR balance in the balance sheet with the master file AR control account balance.
2. Reconcile the AR control figure with the AR subsidiary account total.
3. Select a sample of update entries made to accounts in the AR subsidiary ledger and trace these to transactions in the sales journal (archive file).
4. From these journal entries, identify source documents that can be pulled from their files and verified. If necessary, confirm these source documents by contacting the customers.

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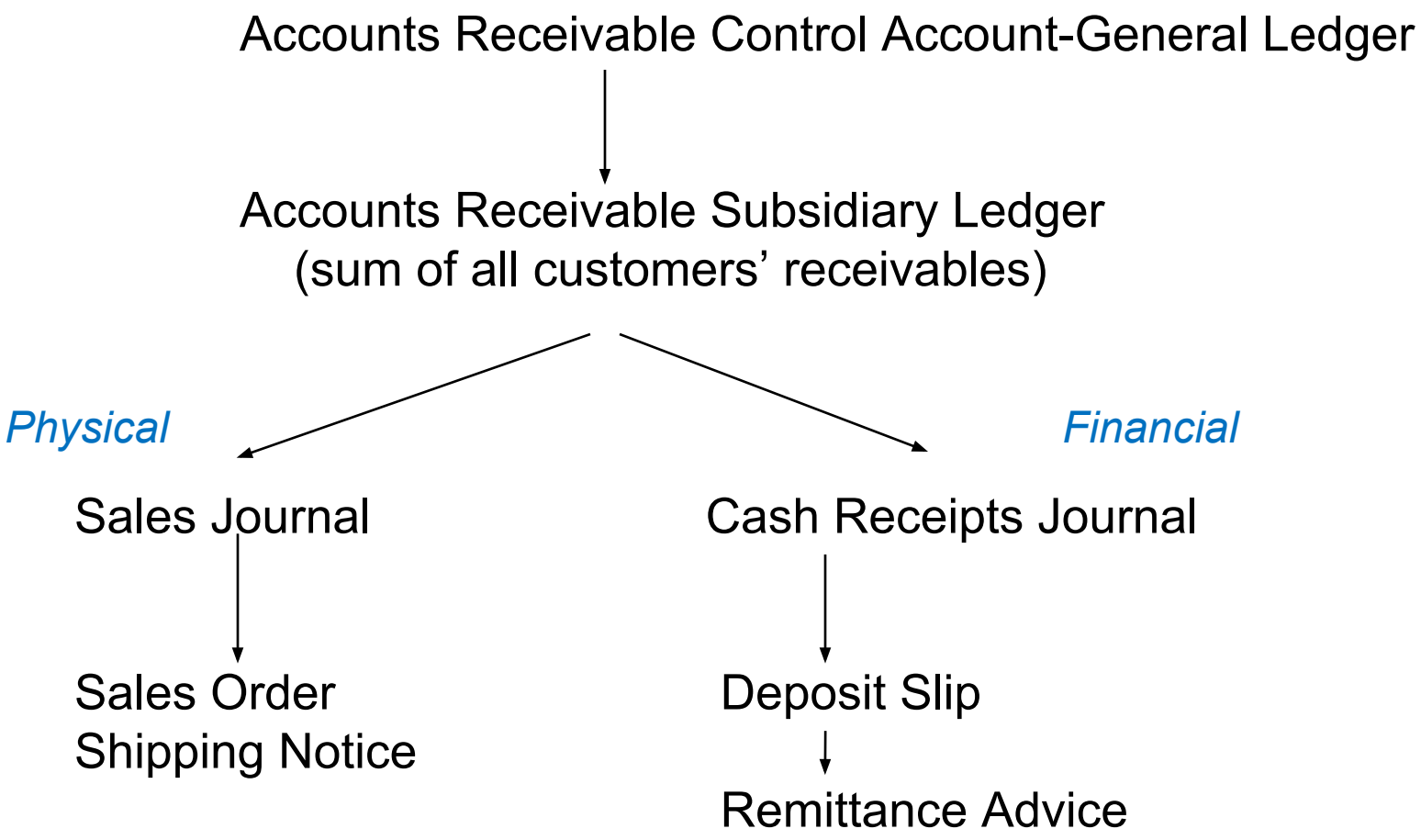
Audit Trail



Accountants should be able to trace in both directions. Sampling and confirmation are two common techniques.

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Example of Tracing an Audit Trail
Verifying Accounts Receivable



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Computer-Based Systems

- The *audit trail is less observable* in computer-based systems than traditional manual systems.
- The data entry and computer programs are the physical trail.
- The data are stored in magnetic files.

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Computer Files

- **Master File** - generally contains account data (e.g., general ledger and subsidiary file)
- **Transaction File** - a temporary file containing transactions since the last update
- **Reference File** - contains relatively constant information used in processing (e.g., tax tables, customer addresses)
- **Archive File** - contains past transactions for reference purposes

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Documentation Techniques

- Documentation in a CB environment is necessary for many reasons.
- Five common documentation techniques:
  - Entity Relationship Diagram
  - Data Flow Diagrams
  - Document Flowcharts
  - System Flowcharts
  - Program Flowcharts

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Entity Relationship Diagram (ERD)

- A documentation technique to represent the relationship between entities in a system.
- The REA model version of ERD is widely used in AIS. REA uses 3 types of entities:
  - **resources** (cash, raw materials)
  - **events** (release of raw materials into the production process)
  - **agents** (inventory control clerk, vendor, production worker)

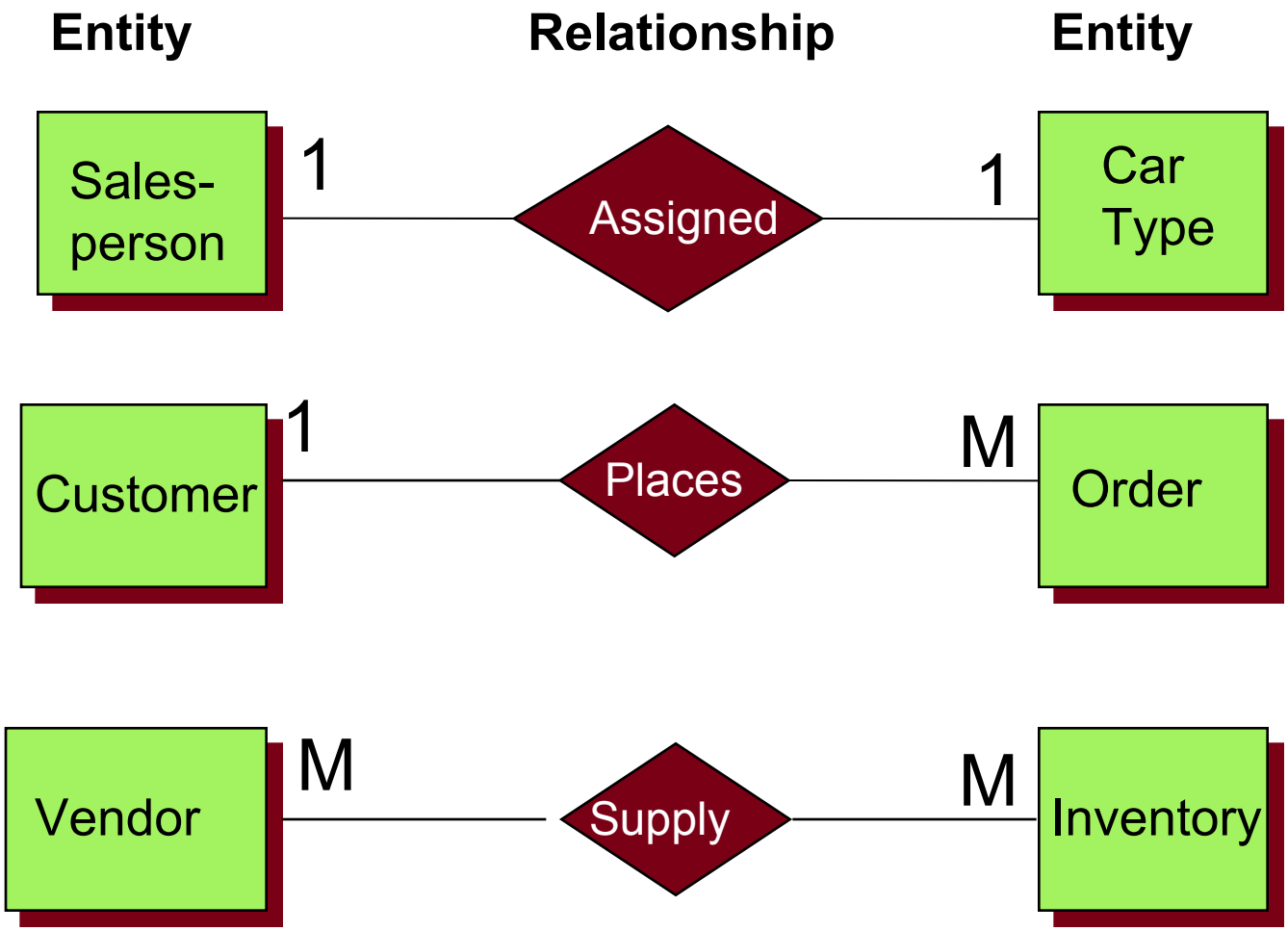
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Cardinalities

- Represent the numerical mapping between entities:
  - one-to-one
  - one-to-many
  - many-to-many

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Cardinalities



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Data Flow Diagrams (DFD)…

- use symbols to represent the processes, data sources, data flows, and entities in a system
- represent the **logical elements** of the system
- do *not* represent the **physical system**

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Data Flow Diagram Symbols

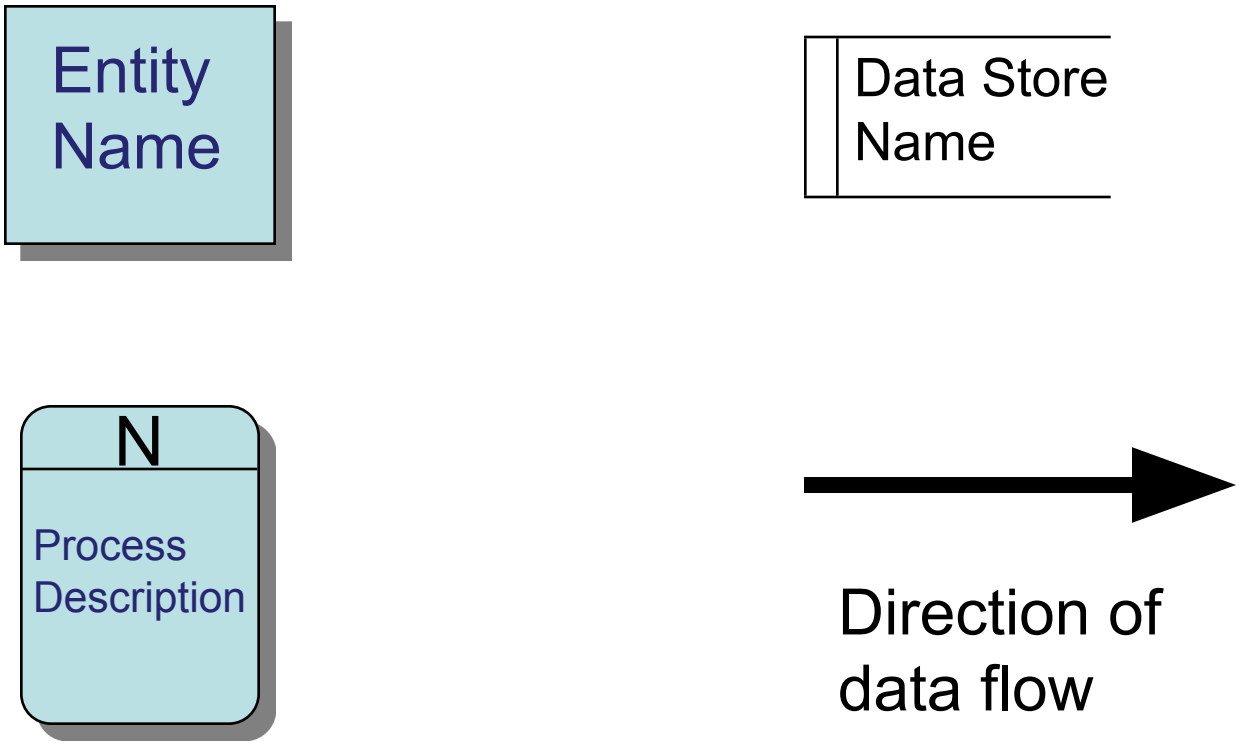


Figure 2-12

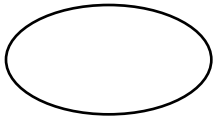
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System Flowcharts…

- illustrate the relationship among processes and the documents that flow between them
- contain more details than data flow diagrams
- clearly depict the separation of functions in a system

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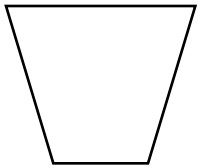
Symbol Set for Representing Manual Procedures



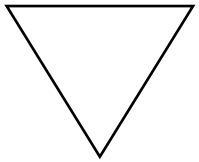
Terminal showing source or destination of documents and reports



Source document or report



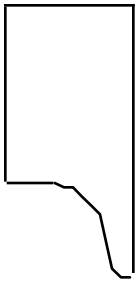
Manual operation



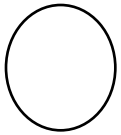
File for storing source documents and reports



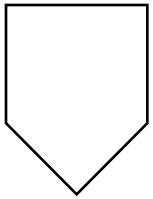
Accounting records (journals, registers, logs, ledgers)



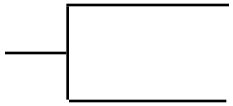
Calculated batch total



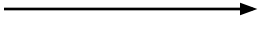
On-page connector



Off-page connector



Description of process or comments



Document flowline

Figure 2-17

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Flowchart Showing Stated Fact I Translated into Visual Symbols

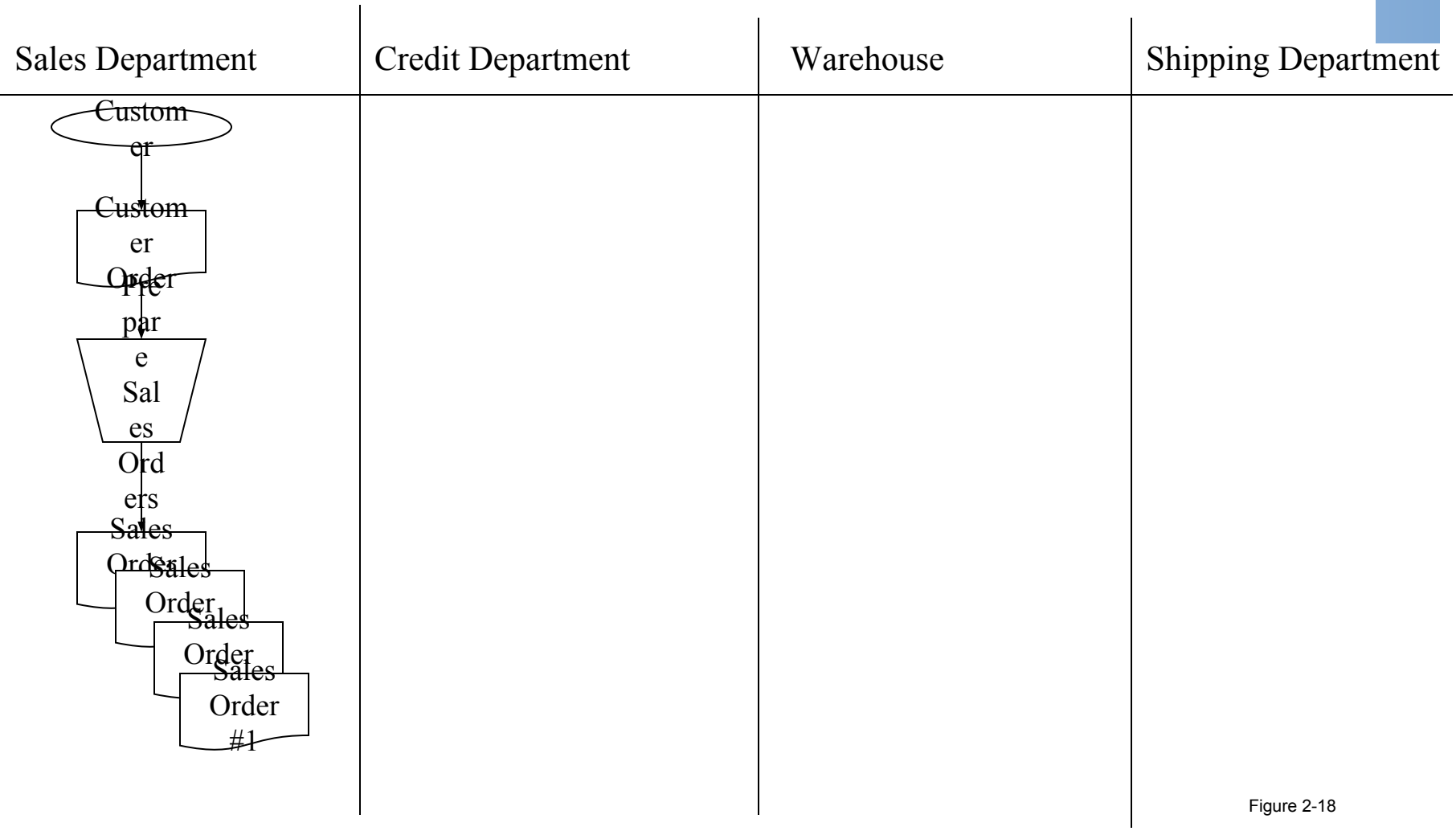


Figure 2-18

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Flowchart Showing All Stated Facts Translated into Visual Symbols

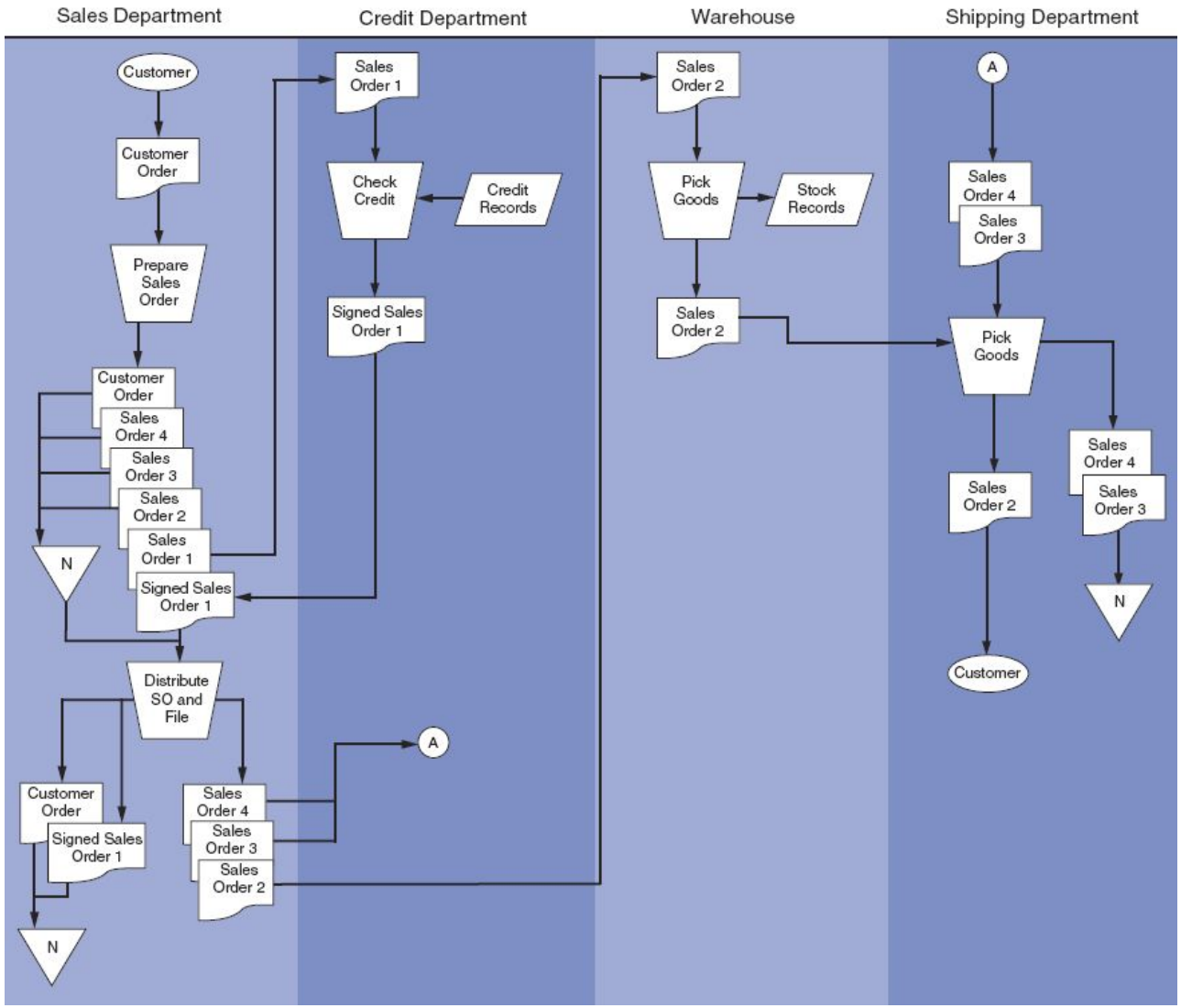


Figure 2-20

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System Flowcharts…

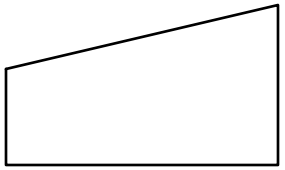
- are used to represent the relationship between the key elements--input sources, programs, and output products--of computer systems
- depict the type of media being used (paper, magnetic tape, magnetic disks, and terminals)
- in practice, not much difference between document and system flowcharts

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Symbol Set for Representing Computer Processes



Hard copy



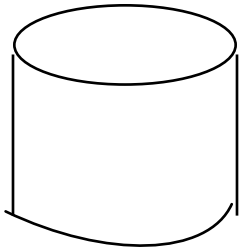
Terminal input/ output device



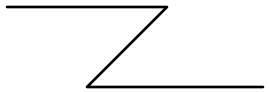
Computer process



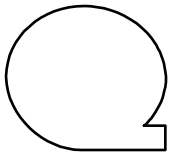
Process flow



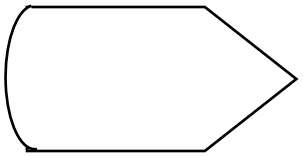
Direct access storage device



Real-time (online) connection



Magnetic tape



Video display device

Figure 2-21

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Flowchart Showing Translation of Facts 1, 2, and 3 into Visual Symbols

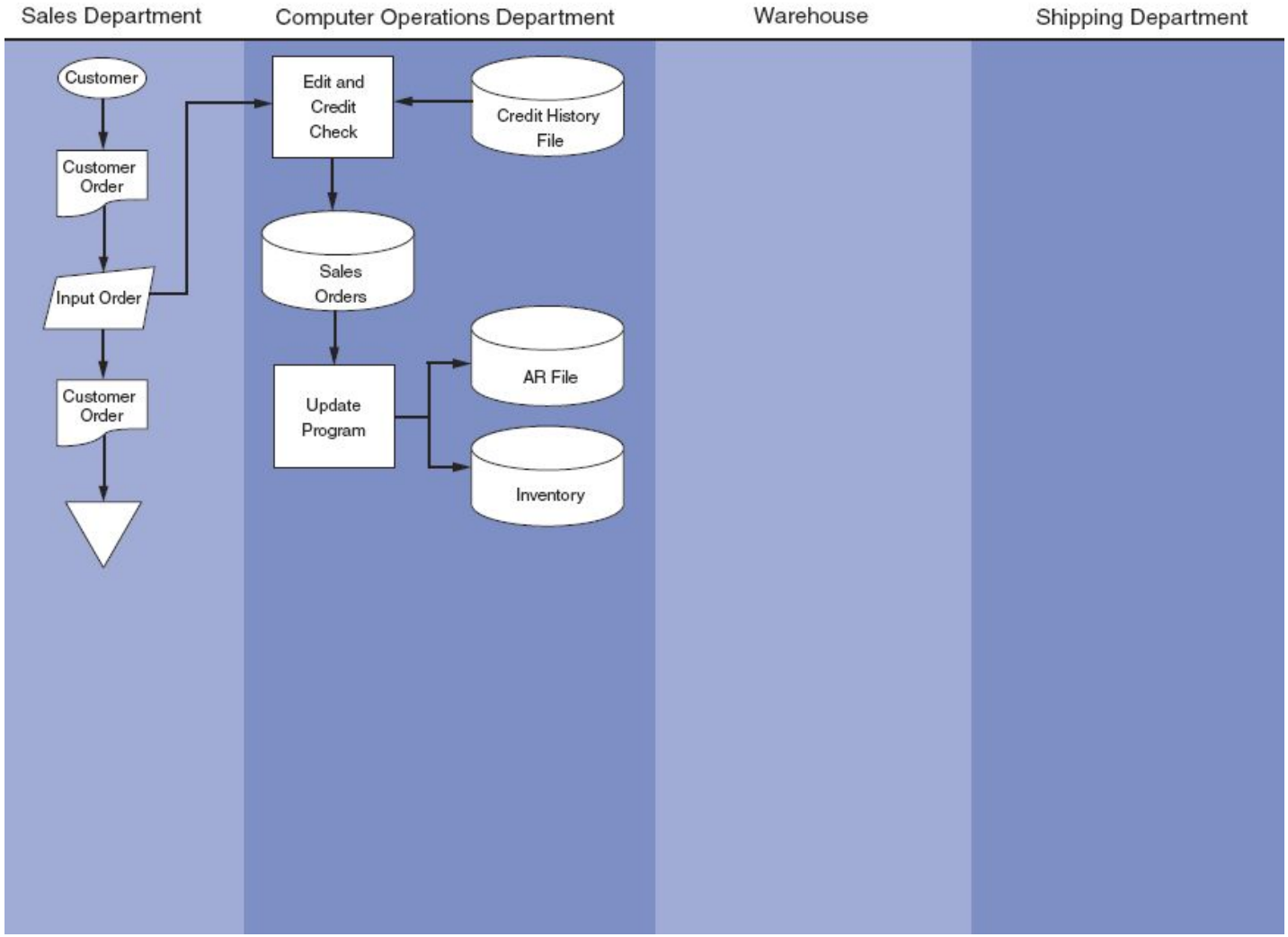


Figure 2-22

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Flowchart Showing All Facts Translated into Visual Symbols

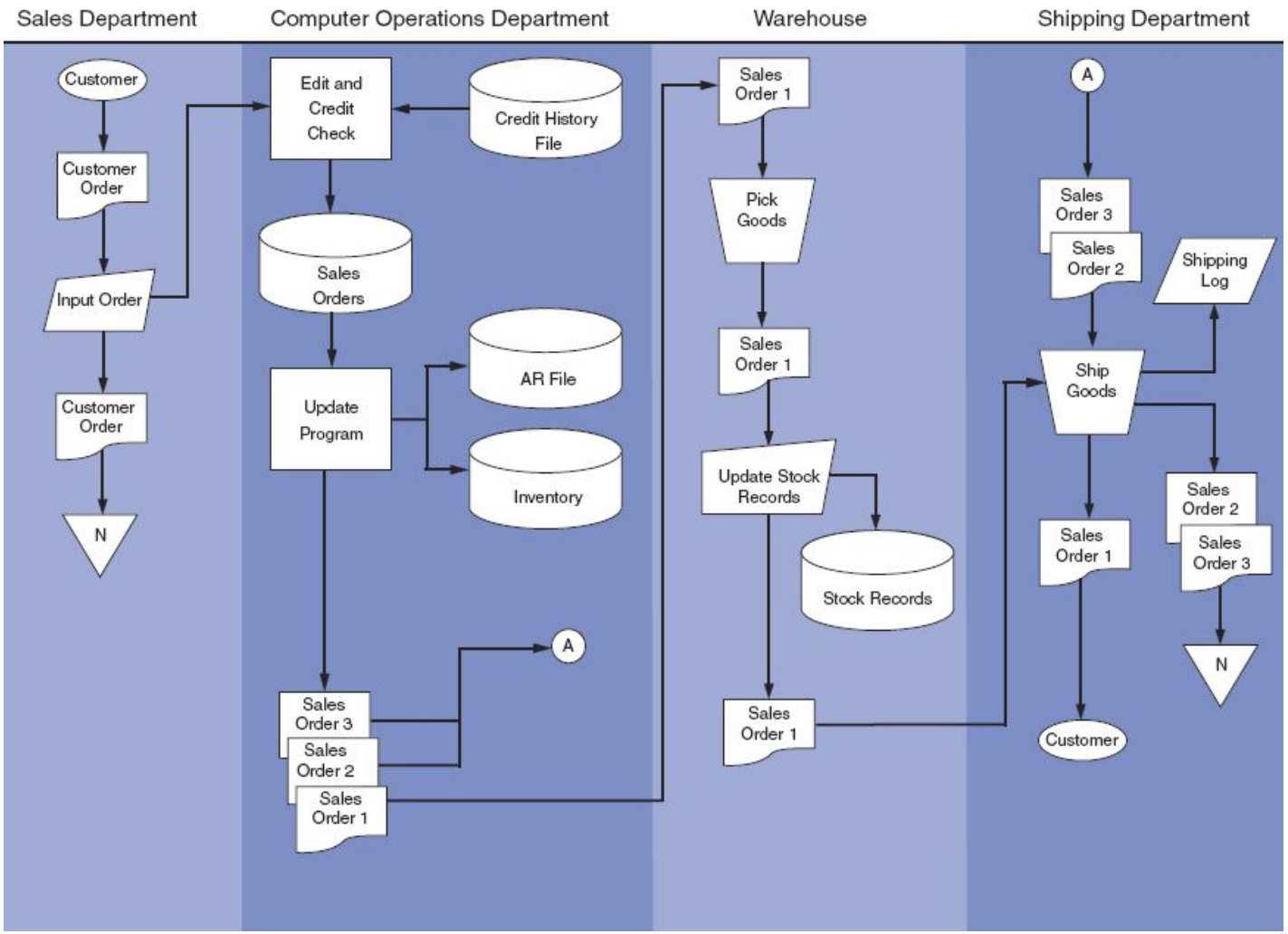


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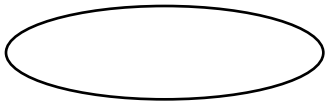
Program Flowcharts…

illustrate the logic used in programs

Program Flowchart Symbols



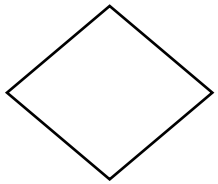
Logical process



Terminal start or end operation



Input/output operation



Decision



Flow of logical process

Figure 2-24

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Modern Systems versus Legacy Systems

- Modern systems characteristics:
  - client-server based and process transactions in real time
  - use relational database tables
  - have high degree of process integration and data sharing
  - some are mainframe based and use batch processing
- Some firms employ legacy systems for certain aspects of their data processing.
  - Accountants need to understand legacy systems.
- Legacy systems characteristics:
  - mainframe-based applications
  - batch oriented
  - early legacy systems use flat files for data storage
  - later legacy systems use hierarchical and network databases
  - data storage systems promote a single-user environment that discourages information integration

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Record Structures for Sales, Inventory, and Accounts Receivable Files

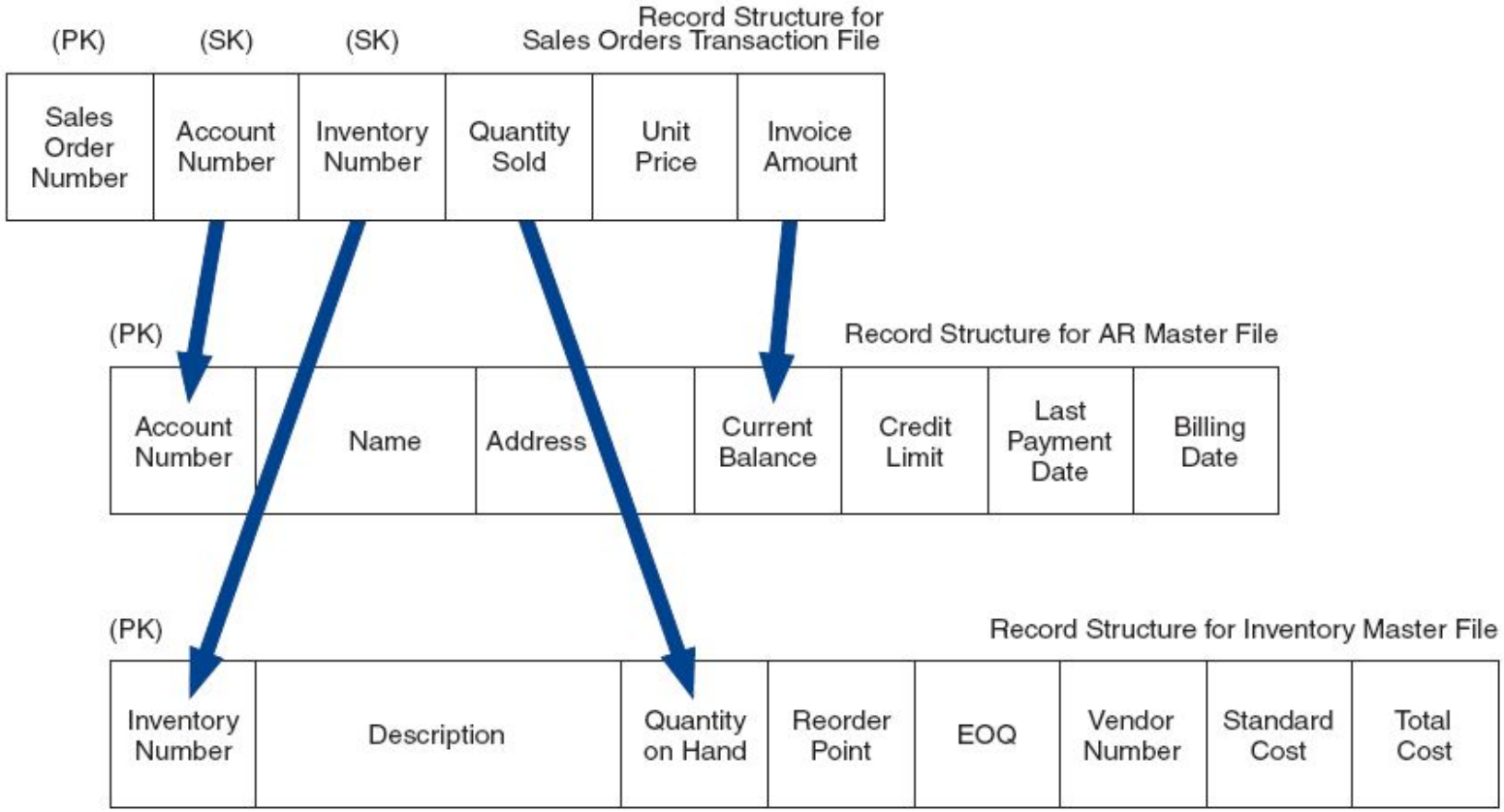


Figure 2-28

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Database Backup Procedures

- Destructive updates leave no backup.
- To preserve adequate records, backup procedures must be implemented, as shown below:
  - The master file being updated is copied as a backup.
  - A recovery program uses the backup to create a pre-update version of the master file.

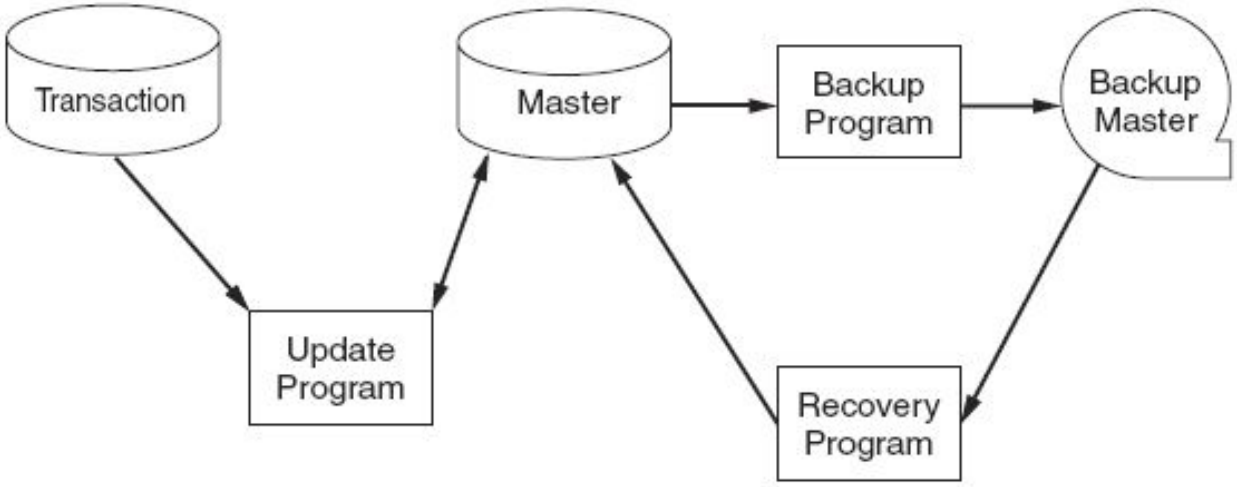


Figure 2-30

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Computer-Based Accounting Systems

- Two broad classes of systems:
  - batch systems
  - real-time systems

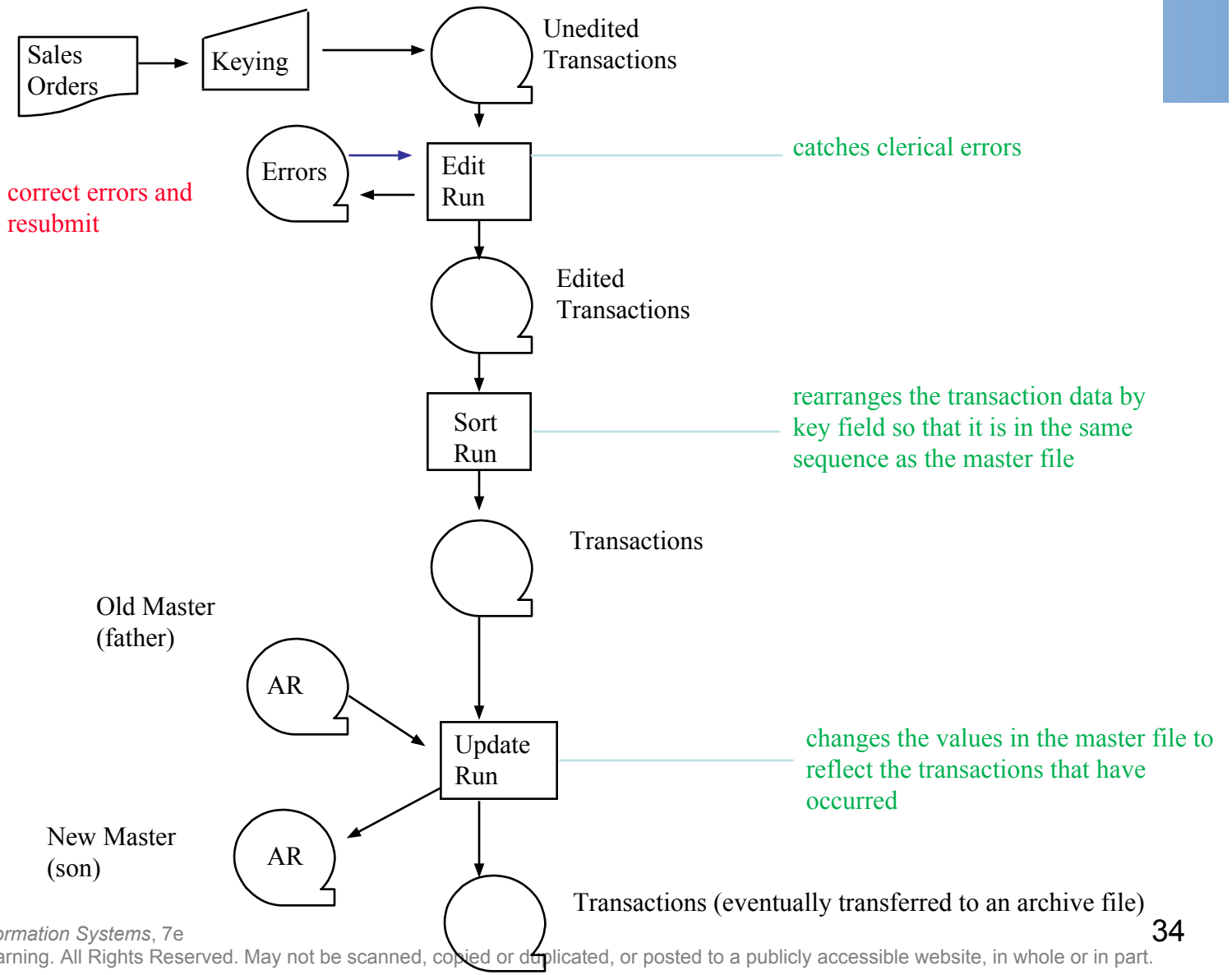
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Batch Processing

- A **batch** is a group of similar transactions that are accumulated over time and then processed together.
- The transactions *must be independent of one another during the time period over which the transactions are accumulated* in order for batch processing to be appropriate.
- A *time lag exists* between the event and the processing.

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Batch Processing/Sequential File



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Steps in Batch Processing/Sequential File

- **Keystroke** - source documents are transcribed by clerks to magnetic tape for processing later
- **Edit Run** - identifies clerical errors in the batch and places them into an error file
- **Sort Run** - places the transaction file in the same order as the master file using a primary key
- **Update Run** - changes the value of appropriate fields in the master file to reflect the transaction
- **Backup Procedure** - the original master continues to exist and a new master file is created

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Advantages of Batch Processing

- Organizations can **increase efficiency** by grouping large numbers of transactions into batches rather than processing each event separately.
- Batch processing **provides control** over the transaction process via control figures.

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Real-Time Systems…

- **process transactions individually** at the moment the economic event occurs
- have **no time lag** between the economic event and the processing
- generally **require greater resources** than batch processing since they require dedicated processing capacity; however, these cost differentials are decreasing
- oftentimes have **longer systems development time**

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Characteristic Differences between Batch and Real-Time Processing

DISTINGUISHING CHARACTERISTIC	DATA PROCESSING METHODS	
	Batch	**Real-Time**
Information Time Frame	Lag exists between time when the economic event occurs and when it is recorded.	Processing takes place when the economic event occurs.
Resources	Generally, fewer resources (hardware, programming, training) are required.	More resources are required than for batch processing.
Operational Efficiency	Certain records are processed after the event to avoid operational delays.	All records pertaining to the event are processed immediately.

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Why Do So Many AIS Use Batch Processing?

- AIS processing is characterized by high-volume, independent transactions, such are recording cash receipts checks received in the mail.
- The processing of such high-volume checks can be done during an off-peak computer time.
- This is one reason why batch processing maybe done using real-time data collection.

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Uses of Coding in AIS

- Concisely represent large amounts of complex information that would otherwise be unmanageable
- Provide a means of accountability over the completeness of the transactions processed
- Identify unique transactions and accounts within a file
- Support the audit function by providing an effective audit trail

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Sequential Codes

- Represent items in sequential order
- Used to prenumber source documents
- Track each transaction processed
- Identify any out-of-sequence documents
- Disadvantages:
  - arbitrary information
  - hard to make changes and insertions

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Block Codes

- Represent whole classes by assigning each class a specific range within the coding scheme
- Used for chart of accounts
  - The basis of the general ledger
- Allows for the easy insertion of new codes within a block
  - Don’t have to reorganize the coding structure
- Disadvantage:
  - arbitrary information

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Group Codes

- Represent complex items or events involving two or more pieces of data using fields with specific meaning
- For example, a coding scheme for tracking sales might be 04-09-476214-99, meaning:

Store Number	Dept. Number	Item Number	Salesperson
04	09	476214	99

- Disadvantages:
  - arbitrary information
  - overused

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Alphabetic Codes

- Used for many of the same purposes as numeric codes
- Can be assigned sequentially or used in block and group coding techniques
- May be used to represent large numbers of items
  - Can represents up to 26 variations per field
- Disadvantage:
  - arbitrary information

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Mnemonic Codes

- Alphabetic characters used as abbreviations, acronyms, and other types of combinations
- Do not require users to memorize the meaning since the code itself is informative – and not arbitrary
  - NY = New York
- Disadvantages:
  - limited usability and availability

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